Chapter 181

Fairlight FX

Fairlight FX are DaVinci Resolve-specific audio plugins that run natively on macOS, Windows, and Linux and provide high-quality audio effects with professional features to all DaVinci Resolve users.

These plugins are available on the Edit and the Fairlight pages and offer various options for repairing faulty audio, creating effects, and simulating spaces.

On the Fairlight page, they're categorized as either Track FX or Audio Effects based on their position in the Fairlight Mixer's signal path.

> For more information on For more information on the Fairlight Mixer's signal path, *see Chapter 177, "Mixing in the Fairlight Page."* This chapter explains what they do and how to use them.

Contents

FAIRLIGHT

Common Controls For All Fairlight FX

Before going into the specific controls of each Fairlight FX plugin, there are some common controls that all plugins share, found at the top of the custom GUI window for each plugin.

Common controls for all Fairlight FX

- **Presets:** A cluster of controls that let you recall and save presets specific to each plugin.

  Add Preset button: Click this button to save the current settings of the Fairlight FX you're using. A dialog box lets you enter a Preset name and click OK.

  Preset dropdown menu: All presets for the currently open plugin appear in this menu.

  Previous/Next preset buttons: These buttons let you browse presets one by one, going up and down the list as you evaluate their effects.

- **A/B Comparison:** A set of buttons that lets you compare two differently adjusted versions of the same plugin. The A and B buttons let you create two sets of adjustments for that plugin, and toggle back and forth to hear which one you like better. The arrow button lets you copy the adjustments from one of these buttons to the other, to save the version you like best while experimenting further.

- **Reset:** A single reset control brings all parameters in the current plugin to their default settings.

  When the Automation is turned on, an automation button appears at the top right of each of the plugins. Automatable parameters for that effect are available in the Plugin dropdown menu for that track.

Automation activated and available to the Fairlight FX

NOTE: Although Default is the preset when initially opened, many of the plugins, have presets already created. Reverb, for instance, has Cathedral, Concert Hall, and Studio presets built in.

These can be excellent starting points to change, rename, and save for your specific needs. These new presets will also be available in the Preset dropdown menu.

TIP: Fairlight FX support “granular” control adjustments by holding down the Option or Shift key while clicking in and dragging left or right within the numerical value fields for virtual sliders in a Fairlight FX plugin or the Inspector.

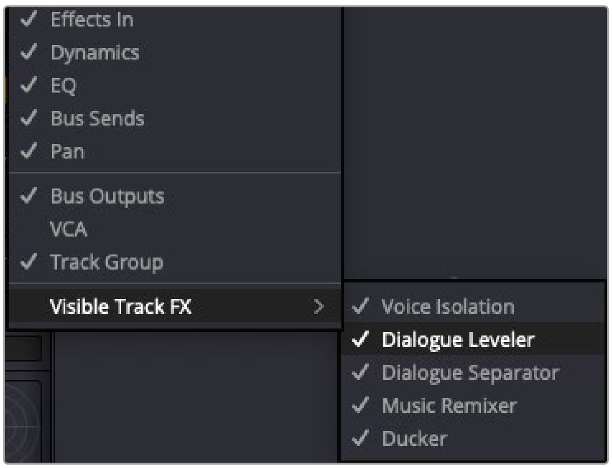
Track FX

These built-in Fairlight FX plugins are accessed via the Track FX section of Audio Track channel strips or in the Inspector. If the Track FX section doesn’t appear in the Mixer, click the three dots in the upper-right corner of the Mixer and select Track FX.

The available plugins in this category are: Voice Isolation, Dialogue Leveler, Dialogue Separator, Music Remixer, and Ducker.

Each plugin occupies a fixed, dedicated insert point in the Track FX section, with the Voice Isolation and Dialogue Leveler visible by default. To reveal the other Track FX plugins, click the three dots in the upper-right corner of the Mixer and select each plugin from the “Visible Track FX” submenu at the bottom.

TIP: Track effects can be applied to stereo-linked groups (stereo-linked tracks).



The Mixer menu

Dialogue Leveler

The Dialogue Leveler analyzes source material to detect dialogue and then "rides down" louder areas, "lifts up" softer areas, and lowers background sounds that are not dialogue.

It works without the typical "pumping" or other unwanted side effects of dynamics processors (compression/limiting) to produce results like those achieved through detailed, manual clip gain adjustments or "riding" the track with fader automation.

The Dialogue Leveler can often "fix and mix" a recording made with a single boom mic, where one character is closer than another or turns away from the mic and levels it out so that the original relative dynamics are preserved while increasing intelligibility and average level.

The Dialogue Leveler plugin

Although Dialogue Leveler can be used on an audio track in real time, it cannot be used on live audio input. It can be used on any mono, stereo, or dual-mono audio track, but it isn't supported on tracks with more than two channels.

Dialogue Leveler controls:

- **Dialogue Leveler:** Enables or disables the plugin.
- **Waveform Processing display:** This shows a scrolling waveform and a gray line indicating Dialogue Leveler processing. However, this display is not available when using the Dialogue Leveler found in the Inspector to process audio clips on the timeline.
- **Preset Menu:** This menu provides the following options:

  Allow wider dynamics: This is the default and is best for audio with a wider-ranging dynamic from loud-to-soft, and where the clip levels are medium to high.

  More lift for low levels: Select this option if the source has more low-level dialogue that you want to boost.

  Lift soft whispery sources: Select this option if the source has whispered dialogue and background noise.

  Optimize moderate levels: Select this option if the source is at medium levels throughout.

- **Reduce Loud Dialogue:** When enabled, louder dialogue is ridden downward when it peaks, acting like a "perfect limiter" where you don't have to adjust the threshold or time parameters. Due to the "near real-time" functionality, analysis occurs prior to audible playback for optimal results.

FAIRLIGHT

- **Lift Soft Dialogue:** This setting finds and lifts low-level dialogue, evening-out material that is quieter while more variable in level. Because the process is dialogue-focused, it doesn't tend to raise background sounds unless they're audible at the same time as the dialogue.

  The Lift Soft Dialogue option is often the most useful of the three options, as it makes quieter lines of dialogue louder and naturally smooth without boosting background noise.
- **Background Reduction:** This option gently removes background noise based on the internal preset selected in the Preset menu.
- **Output Gain:** You can use this control to adjust the output level by clicking and dragging it or entering a value in the numeric field. The range of the Output Gain control of 0.0 to +6.0 dB.

TIP: Before using the Dialogue Leveler on a track, it can be useful to set an optimum baseline level for use with the default "Allow wider dynamics" preset by normalizing the audio.

For more information on normalizing audio, *see Chapter 176, "The Fairlight Inspector."*

Dialogue Separator (Studio Version Only)

Dialogue Separator uses DaVinci Neural Engine AI to give you individual control over the level of dialogue, background sounds, and the reverberant field or ambient room sound ("ambience").

The Dialogue Separator plugin

The Dialogue Separator is useful when, for example, a source file has a great roomy sound, but there's music or crowd noise in the background. In this instance, you can rebalance the audio to reduce the background sounds or adjust the room sound to make the dialogue more intelligible.

Another use case would be when someone moves toward the camera in a roomy environment while speaking. You could use automation to rebalance the audio so that the ambient room sound gradually becomes "drier" as they get closer, without affecting the background.

Dialogue Separator is currently a mono-only plugin. If your source is stereo, you'll need to split it into dual-mono and process both mono tracks separately.

Ducker

This plugin lets you use the audible signal from a track or multiple tracks to lower the level of another.

Although it can be used in many creative ways, a common use of “ducking” is automatically lowering a music or sound effects bed so that dialogue or a VO track can be heard more prominently in a mix.

This is achieved without compressing the incoming signals. When set up correctly in the Ducker interface or via the Audio tab in the Inspector, your audio tracks can practically “mix themselves.”

In a simple use case, the Ducker uses level changes of a single incoming dialogue or VO track to decrease the volume of the sound effects bed of a noisy freeway.

In a more complex situation, such as a scene where three characters are conversing next to the same noisy freeway, a single Ducker plugin can use the level changes of each dialogue track to attenuate the traffic noise.

With only one input source (Dialogue 1), the Ducker would only be effective when the first character speaks. However, this new feature lets you add the other two dialogue tracks (Dialogue 2 and Dialogue 3) as input sources for attenuating the traffic noise.

The Ducker plugin

Using the Ducker

Continuing with the use cases mentioned above:

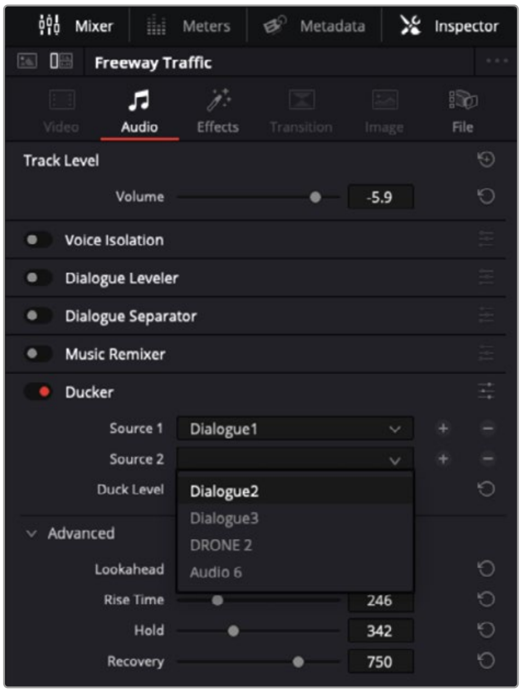
1. Choose the target track that you want to attenuate (the Freeway Traffic track), and open the Ducker.
2. Click the Source dropdown and select Dialogue 1 as the input source.
    - To use additional input sources, Command-click the other track names; in this example, Dialogue2 and Dialogue3.
    - Play a bit of your sequence, and adjust the duck level and other controls to taste.

Adding Input Sources via the Inspector

If you prefer working within the Inspector:

1. Select the Freeway Traffic track, ensure the Ducker is switched on, and open the Inspector’s Audio tab.
2. Click the plus sign to the right of the Source 1 dropdown containing Dialogue 1 to create a dropdown menu for Source 2, then choose Dialogue2 from the list.

3 Click the plus sign to the right of the Source 2 dropdown to create a dropdown menu for Source 3, then choose Dialogue3 from the list.



Adding Ducker input sources in the Inspector

Ducker Controls

- **Duck Amount:** This control determines level attenuation on the target track in dB with a default value of 2.7 dB. Most of the time, a value between 2.0 dB and 5.0 dB works the best.
- **Lookahead:** Adjusts the amount of lookahead time in milliseconds, with a default value of 15ms. Lookahead allows the Ducker to start attenuating the destination track before the source track is audible.

  The larger the value, the more the Ducker will “pre-anticipate” the source track. If you want the Ducker to start lowering the sound before a line of dialogue or VO begins, you can experiment with higher values, but the default will usually produce good results.
- **Rise Time:** This determines how many milliseconds it takes for the Ducker to reach the amount of reduction specified by the Duck Amount. The default value is 10 milliseconds, which should work for most situations. Longer Rise Times may be useful with higher Duck Amount values and longer gaps in the source track.
- **Hold:** This parameter determines how long the target signal will be attenuated. The default is 150ms.
- **Recovery Time:** This value indicates how quickly the target signal will return to its previous level after attenuation. The default time value is 750ms, but setting the correct Recovery time is crucial to a natural-sounding ducking process, so you should try different values to find the best results.

Ducker Graph Display

This area shows a gray line representing the incoming signal against the action of the Ducker, shown in yellow. Dips in the yellow line signify attenuation of the target track based on the Duck Amount in combination with the other parameters.

Music Remixer (Studio Version Only)

This DaVinci Neural Engine AI-based plugin splits music into individual stems: Vocals, Drums, Bass, Guitar, and "Other" ("everything else"), letting you creatively rebalance or remix the track.

You can use the level controls to adjust the volume of each stem and the Mute buttons to remove or bring them back into your mix. For example, you may want to remove or lower a vocal because it clashes with a featured piece of dialogue.

The Music Remixer plugin

Voice Isolation (Studio Version Only)

Voice Isolation uses an AI model trained for any type of human voice, male or female, young or old, which lets you completely remove loud, undesirable sounds from existing voice recordings with incredible results.

The Voice Isolation plugin

You can get rid of all kinds of background noise, from air conditioners or fans to jackhammers, restaurant background noise, music playing during a featured piece of dialogue, and more.

The Amount control adjusts the mix between the original source and the isolated voice. When set to 50, the mix is roughly equal. Values between 70 and 80 work well for natural results while strongly isolating the source.

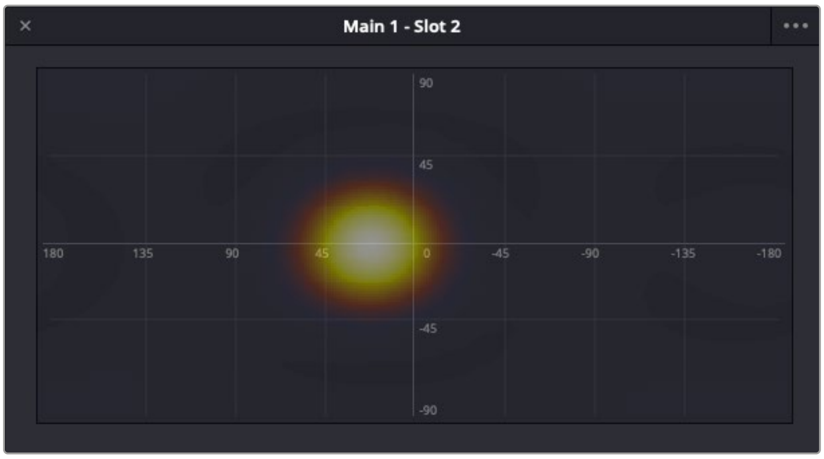
Although Voice Isolation can be used on an audio track in real time, it cannot be used on live audio input. It can be used on any mono, stereo, or dual-mono audio track, but it isn't supported on tracks with more than two channels.

Insert Effects

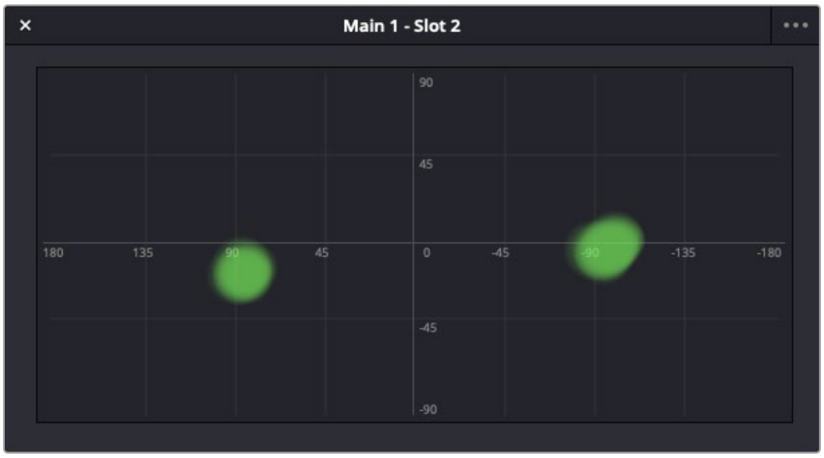
Ambisonics Meter

This Fairlight FX plugin can be inserted on an Ambisonics track or bus to present a graphic display of your Ambisonic sound field. Clicking the three dots in the upper right opens an options menu where you can choose either "power" (Power Map), or "Sonar" style map as the metering method.

Both meters convey the intensity of the sound and its current location in the Viewer.



Ambisonics meter - Power Map view



Ambisonics meter - Sonar view

FAIRLIGHT

Chain FX

Fairlight Chain FX lets you build chains of up to six Fairlight FX or compatible Audio Units or VST effects plugins, which can be adjusted using a single control.

Chain FX plugin

Master Controls

The Chain FX window has the following controls located in the upper section of the interface.

Preset Menu: This is where you can save and recall Chain FX settings.

- Chain FX combinations can be saved as presets by clicking the Plus button to the left of the Preset menu and recalled by clicking the menu and making a selection.
- You can also scroll through the presets by clicking the left or right "arrow" (< or >) to the right of the drop-down menu.
- You can save a Chain FX configuration as the default by selecting the Set as Default option from the Options menu (three dots) located in the upper-right corner, to the right of the Reset button.
- Chain FX presets can be copied and pasted to other instances of the plugin for a faster workflow.

TIP: Chain FX presets can also be recalled by clicking the plus sign on an insert slot, followed by Chain FX.

Compare Buttons (A and B): These buttons let you quickly compare two different Chain FX settings. If you create a Chain FX setting or select a preset when button A is active (red), then click button B and create a different setting or choose a different preset, clicking between the buttons quickly switches from one to the other.

Bypass Button: Switches the Fairlight EQ on and off.

Reset Button: Clicking the circular Reset icon, located in the upper-right corner of the EQ, undoes the most recent action you've taken in the plugin interface.

Sidechain (SC) Section:

- **On Button:** Clicking this button activates the Sidechain section of the plugin, prompting you to select an audio track or buss as the input source from the Source drop-down menu.

> For more information about Sidechaining, *see Chapter 180, "Audio Effects."*

- **Listen Button:** Clicking this button lets you hear only the incoming audio signal, which is helpful if you need to confirm that the correct audio signal is being sent to the Sidechain module.
- **Source:** This button selects the track or buss the incoming audio signal is coming from.

Building an FX Chain

To create an effect chain in a Chain FX instance:

1 1. Open an empty Chain FX plugin by clicking an empty Effects slot on a Mixer Channel Strip and choosing Multi Effect > Fairlight FX > Chain FX.

   Ensure that the plugin is active.

2 Add the first Fairlight FX or compatible third-party plugin by clicking the plus sign in the Plugin Chain section and selecting a plugin from the menu.

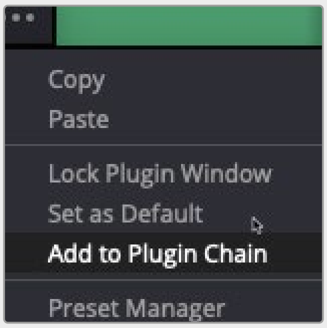
   While the plugin is active, either recall an existing preset or adjust the controls to dial in your desired settings.

3 Repeat the previous step until you have built an FX chain of up to six plugins that works for you.

4 Save the resulting combination of plugins, in their current state as a Chain FX Preset, by clicking the Plus button to the left of the Preset menu.

> **TIP:** Clicking the name of any plugin listed in the effect chain opens its respective plugin window.

You can also create a new effects chain by opening any compatible plugin, such as a compressor, clicking the three dots in the upper-right corner of the plugin window, and selecting Add to Plugin Chain from the menu.



Add to Plugin Chain menu option

This opens a new Chain FX instance with the plugin you originally selected (the compressor, in this example) already inserted in the first slot.

- The Chain FX plugin replaces the compressor in the mixer channel insert.
- You can add more plugins to the effect chain by clicking the plus sign in the Plugin Chain section and selecting a plugin from the menu.

Chorus

A classic Chorus effect, this layers voices or sounds against modulated versions of themselves to add harmonic interest in different ways.

An animated graph shows the results of adjusting the Modulation parameters of this plugin, giving you a visualization of the kind of warble or tremolo that will be added to the signal as you make adjustments.

The Chorus Fairlight FX

Chorus has the following controls:

- **Bypass:** Toggles this plugin on and off.
- **Input Format:** (Only visible when Chorus is inserted on a multi-channel track.) Lets you choose how multiple channels are input to the Chorus. Stereo sets separate Left and Right channels. Mono sums Left and Right to both channels. Left inputs the Left channel only, and Right inputs the Right channel only.
- **Delay:** The amount of delay between the original sound and the Chorus effect.

  Delay Time: Length of the Chorus delay lines.

  Separation: Time separation of the delay voices.

  Expansion: Sets L/R length differences, phase offset of modulators.

- **Modulation:** These controls adjust the low frequency oscillator (LFO) that drives the tremolo of the chorus effect in different ways.

  Waveform: Specifies the shape of the LFO that modulates the rate of the Chorus, affecting the timing of the oscillations. There are six options for the oscillator for you to choose from:

   **Sine** (Smooth oscillations)

   **Triangle** (Sudden oscillations)

   **Saw1** (Jerky oscillations)

   **Saw2** (Jerky oscillations)

   **Square** (Hard stops between oscillations)

   **Random** (Randomly variable oscillations)

  Frequency: Rate of LFO controlling the Chorus. Lower values generate a warble, higher values create a tremolo.

  Pitch: Amount of frequency modulation, which affects the pitch of the Chorus.

  Level: Depth of level modulation. Affects the "length" of the segment of Chorus that's added to the sound. Low values add only the very beginning of the Chorus effect, high values add more fully developed Chorus warble or tremolo.

- **Feedback:** A signal fed back to the Chorus Delay Line

  Amount (%): The percentage of signal fed back to the Chorus Delay Line. Values can be positive or negative, the default is 0 (no effect). Increasing this parameter adds more of the Chorus effect to the signal, lowering this parameter adds more of the inverted Chorus effect to the signal. At values closer to 0, only a faint bit of Chorus can be heard in the audio, but at values farther away from 0 (maxing at +/- 99), a gradually pronounced Chorus becomes audible.

  Blend (%): Amount of feedback which bleeds into the opposite channel (Stereo mode only).

- **Output:** Controls for adjusting the final output from this plugin.

  Dry/Wet (%): A percentage control of the output mix of "dry" or original signal to "wet" or processed signal. 0 is completely dry, 100% is completely wet.

  Output Level (dB): Adjusts the overall output level of the affected sound.

De-Esser

A repair plugin specific to dialog. The De-Esser is a specialized filter that's designed to reduce excessive sibilance, such as hissing "s" sounds or sharp "ts" sounds, in dialogue or vocals.

A graph shows you which part of the signal the controls are set up to adjust, while reduction and output meters let you see which part of the signal is affected and what level is being output.

The De-Esser Fairlight FX

The De-Esser has the following controls:

- **Bypass:** Toggles this plugin on and off.
- **Listen to Ess Only:** This checkbox at the top right lets you listen only to the ess that is being removed. This is very useful to determine if too much signal is being removed or if more ess attenuation can be applied.
- **Frequency Range:** Two controls let you target the frequency of the "s" sound for a particular speaker.

  Target Frequency: A knob that lets you target the frequency of the offending sibilance. Sibilant sounds are usually found in the range of 5 - 8kHz.

  Range: switches the operational mode of the De-Esser. Three choices (from top to bottom) let you switch among Narrow Band, Wide Band, and All High Frequency which processes all audio above the source frequency.
- **Amount:** Adjusts the amount de-essing that's applied.
- **Reaction Time:** Adjusts how suddenly de-essing is applied. There are three choices.

  Slow: Equivalent to a slow attack.

  Medium: Equivalent to a medium attack.

  Fast: Equivalent to a fast attack.

De-Hummer

A repair plugin with general applications to any recording. Eliminates hum noise that often stems from electrical interference with audio equipment due to improper cabling or grounding. Typically 50 or 60 cycle hum is a harmonic noise, consisting of a fundamental frequency and subsequent partial harmonics starting at twice this fundamental frequency.

A graph lets you see the frequency and harmonics being targeted as you adjust this plugin's controls.

The De-Hummer Fairlight FX

De-Hummer has the following controls:

- **Bypass:** Toggles this plugin on and off.
- **Listen to Hum only:** This checkbox at the top right lets you listen only to the hum that is being removed. This is very useful to determine if too much signal is being removed or if more hum attenuation can be applied.
- **Frequency:** Target source fundamental frequency. A knob lets you make a variable frequency selection, while radio buttons let you select common frequencies that correspond to 50Hz/60Hz electrical mains that are the typical culprits for causing hum.
- **Amount:** Adjusts how much De-Hum extraction you want to apply.
- **Slope:** Adjusts the ratio of fundamental frequency to partial harmonics, the adjustment of which lets makes it possible for various kinds of hum to be targeted. For example, a value of 0 biases hum extraction towards the fundamental frequency, while a value of 0.5 gives equal extraction of all harmonics (up to 4), and finally a value of 1.0 targets the higher frequency partials.

Delay

An effects plugin. A general purpose stereo delay effect, suitable for tasks varying from track doubling, to early reflection generation, through simple harmonic enhancement. Processes in stereo or mono, depending on the track it's applied to.

A graph shows the timing and intensity of the echoes generated by this plugin on each channel, and an Output meter displays the output level of the resulting signal.

The Delay Fairlight FX

Delay has the following controls:

- **Bypass:** Toggles this plugin on and off.
- **Input Format:** (Only visible when Delay is inserted on a multi-channel track.) Lets you choose how multiple channels are input to the delay. Stereo sets separate Left and Right channels. Mono sums Left and Right to both channels. Left inputs the Left channel only, and Right inputs the Right channel only.
- **Filters:** Alters the proportion of frequencies that are included in the delay effect. When the Delay plugin is inserted on a Mono Channel, the Left and Right sections are replaced with a single "Delay" section.

  Low Cut (Hz): A global High Pass filter.

  High Cut (Hz): A global Low Pass filter.

- **Delay:** Adjusts the timing of the delay.

  Left/Right Delay (ms): Delay time of each channel.

  Left/Right Feedback (%): Feedback % of the Left or Right channel back to itself. A negative value equates to % of feedback with a phase reverse from the original signal.

- **Feedback:** Controls for adjusting the amount of bleed between channels.

  High Ratio: Adjusts the frequency of a damping filter for the feedback signal.

  Stereo Blend: Adjusts the proportion of signal from Left and Right channel feedback which feeds into the opposite channel. When the Delay plugin is inserted on a Mono channel, Stereo Blend control does not appear.

- **Output:** Controls for adjusting the final output from this plugin.

  Dry/Wet (%): A percentage control of the output mix of "dry" or original signal to "wet" or processed signal. 0 is completely dry, 100% is completely wet.

  Output Level (dB): Adjusts the overall output level of the affected sound.

Dialogue Processor

The Dialogue Processor chains together six different common audio processing operations inside of a single plugin, each tuned to the specific needs of adjustments to improve recorded dialog. The specialized De-Rumble, De-Pop, De-Ess, Comp(ressor), Expander, and Excite controls each have a streamlined set of controls tailored to the specific types of common adjustments that may be required for a great sounding dialogue track.

The Dialogue Processor plugin

- **De-Rumble:** The de-rumble filter is a high pass filter that allows you to remove subsonic rumble (subharmonic low frequencies) in your vocal microphone signal. A safe value to use for most condenser microphones is 40 or 50Hz. Using higher values can start to remove lower frequencies in the source signal, which can be a useful creative effect at times.

  Frequency: Sets the high-pass filter to frequencies between 40Hz to 235Hz.

- **De-Pop:** Controls the center frequency for a specialized filter for removal of vocal "popping" sounds, or plosives, that can occur when someone speaks close to a microphone capsule on-axis.

  Frequency: Sets the filter to frequencies between 50Hz to 200Hz.

  Amount: Sets the amount of the filter effect on the signal with lower values effecting less and higher values effecting more. A meter next to the amount knob shows how the signal is being effected.

- **De-Ess:** Sets the center point of a specialized bandpass filter for removal of vocal "Esses," which can be overly present due to sibilance. De-essing allows you to raise average level and brighten a vocal source without it sounding harsh, as our ears are very sensitive to overly present sibilance. The best frequency range for male and female voices tends to be between 6 and 7 kHz.

  Frequency: Sets the filter to frequencies between 700Hz to 9000Hz.

  Amount: Sets the amount of the filter effect on the signal with lower values effecting less and higher values effecting more. A meter next to the amount knob shows how the signal is being effected.

- **Compressor:** Used to compress a signal's dynamic range by reducing differences in level between the loudest and quietest parts of the input signal. Compression can be used to raise the signal's overall level, boosting it without clipping and thus increasing perceived loudness. Compression is often used to give voices more presence within a mix and to smooth out changes in the levels of tracks that may have too much dynamic range for its context.

  Threshold (dB): Sets the level at which the signal will start to be effected from –40dB to –8dB.

  Amount: Sets the amount of the compression on the signal with lower values effecting less and higher values effecting more. A meter next to the amount knob shows how the signal is being effected.

  Fast/Slow: Determines the speed at which the effect is applied, fast or slow.

- **Expander:** Emphasizes differences in volume by lowering the level of soft parts of the signal relative to the level of louder parts and can be used to minimize noise while increasing the dynamic range of a signal.

  Threshold (dB): Sets the level at which the signal will start to be effected from –40dB to –8dB.

  Range: Amount of decrease in signal level in dB, as expansion lowers the softer parts of the signal.

  Fast/Slow: Determines the speed at which the effect is applied, fast or slow.

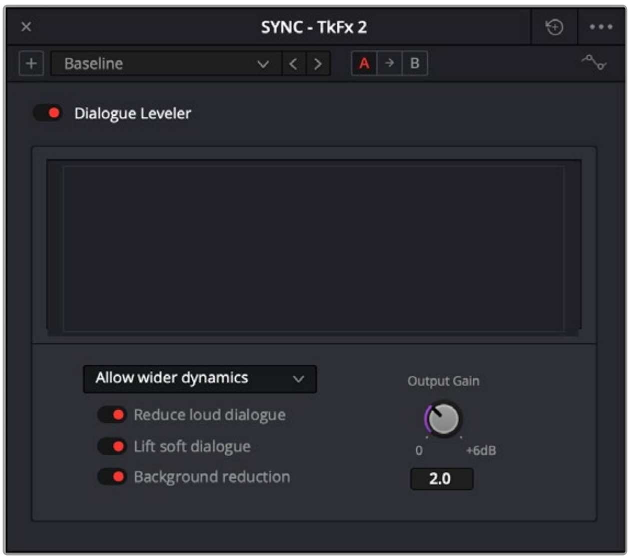
- **Excite:** Synthesizes additional high frequency content in a source signal, using various techniques. Used subtly, it can add more presence to sounds that may be overly dull without having to use EQ.

  Amount: Sets the amount of the exciter on the signal with lower values effecting less and higher values effecting more. A meter next to the amount knob shows how the signal is being effected.

  Female/Male: Tailors the exciter's response to be optimized for a male or female voice.

Dialogue Leveler

The Dialogue Leveler analyzes source material to detect dialogue, and then "ride down" louder areas, "lift up" softer areas, and lower background sounds that are not dialogue. It works without typical dynamics processor "pumping" (compression/limiting) or other obvious side effects, and produces results similar to detailed manual clip gain adjustments or by carefully "riding" the track with fader automation. Dialogue Leveler can often "fix and mix" a recording with a single boom mic where one character is closer than another or turns away from the mic at times, and level it out so that the original relative dynamics are preserved while increasing intelligibility and average level.



The Dialogue Leveler plugin

Dialogue Leveler can be used on track audio in real time but not on live audio input. Dialogue Leveler can be used on any mono or stereo (or multi-mono) audio track, but it is not supported on greater-than-stereo tracks.

Dialogue Leveler has the following controls:

- **Dialogue Leveler:** Enables or disables Dialogue Leveler processing.
- **Waveform Processing display:** Shows a scrolling waveform display and indicates Dialogue Leveler processing with a gray line. However, note that the display does not appear in the version of Dialogue Leveler for clip-based processing.
- **Preset Menu:** The preset menu provides the following options:

  Allow wider dynamics: This is the default, and is best for sources with wider ranging dynamic levels from loud-to-soft, and where the clip levels are medium to high.

  More lift for low levels: Select this option if the source has more low level dialogue that you want to boost.

  Lift soft whispery sources: Select this option if the source has whispered dialogue and background noise.

  Optimize moderate levels: Select this option if the source is at medium levels throughout.

- **Reduce Loud Dialogue:** When enabled, louder dialogue is ridden downward on peaks and acts somewhat like a "perfect limiter" where you don't have to adjust threshold or time constants. Due to the "near real time" aspect, analysis occurs prior to audible playback for optimal results.
- **Lift Soft Dialogue:** When enabled, finds low level dialogue and lifts and evens out material that is more variable in level and softer, but because the process is dialogue focused, it doesn't tend to raise background sounds (unless they are happening at the same time as the dialogue itself). More often than not, the Lift Soft Dialogue option is the most useful of the three options, as it makes less audible lines of dialogue more audible and naturally smooth, while not boosting background noise.
- **Background Reduction:** When enabled, reduces background sounds by focusing on dialogue and gently removing them based on the internal presets (Preset menu).
- **Output Gain:** Adjust the Output Gain by clicking and dragging on the Output Gain control or by entering a value in the numeric field (Output Gain control in dB with 0 to +6 dB range with .1dB resolution).

Distortion

An effects plugin. Creates audio distortion that's useful for sound design and effects, ranging from simple harmonic distortion simulating an audio signal going through primitive or faulty electronics (such as bad speakers, old telephones, or obsolete recording technologies), all the way to mimicking an overdriven signal experiencing different intensities of hard clipping (think someone yelling through a cheap bullhorn, megaphone, or PA system). This plugin includes soft tube emulation in the output stage.

An animated graph shows the results of adjusting the Distortion parameters of this plugin, giving you a visualization of the kind of harmonic distortion, waveshaping, and clipping that will be modifying the signal as you make adjustments. Input and Output meters let you see how the levels are being affected.

The Distortion Fairlight FX

Distortion has the following controls:

- **Bypass:** Toggles this plugin on and off.
- **Filters:** Two filters let you simulate devices reproducing limited frequency ranges.

  LF Cut: Low frequency distortion shaping.

  HF Cut: High frequency distortion shaping.

- **Distortion:** Three sets of controls let you create the type and intensity of distortion you want.

  Mode buttons: Switch the operational mode of distortion. The one to the left, Distortion, creates harmonic distortion. The button to the right, Destroy, is a more extreme polynomial waveshaper.

  Distortion: Adjusts the amount of distortion that's applied to the signal. Higher values distort more.

  Ceiling: Adjusts the level of input signal that triggers clipping.

- **Output:** Controls for adjusting the final output from this plugin.

  Dry/Wet (%): A percentage control of the output mix of "dry" or original signal to "wet" or processed signal. 0 is completely dry, 100% is completely wet.

  Output Level (dB): Adjusts the overall output level of the affected sound.

  Auto Level button: Applies automatic compensation for gain added to the signal due to the distortion being applied. Having this button turned on prevents the signal from becoming dramatically and unexpectedly increased, while turning it off frees you to do what you want, if what you want is to hear a lot of distortion.

Echo

An effects plugin. A classic Echo effect, simulating the reflection of sounds that occur with a delay after the direct sound is heard. Processes in stereo or mono, depending on the track it's applied to.

A graph shows the timing and intensity of the echoes generated by this plugin on each channel, and an Output meter displays the output level of the resulting signal.

The stereo Echo Fairlight FX

Echo has the following controls:

- **Bypass:** Toggles this plugin on and off.
- **Input Format:** (Only visible when Echo is inserted on a multi-channel track.) Lets you choose how multiple channels are input to the echo. Stereo sets separate Left and Right channels. Mono sums Left and Right to both channels. Left inputs the Left channel only, and Right inputs the Right channel only.
- **Filter:** Alters the proportion of frequencies that are included in the delay effect. When the Delay plugin is inserted on a Mono Channel, the Left and Right sections are replaced with a single "Delay" section.

  Low Cut (Hz): A global High Pass filter.

  High Cut (Hz): A global Low Pass filter.

  Feedback: Adjusts the frequency of a damping filter for the feedback signal.

- **Left Channel:** Parameters that independently affect delay on the Left Channel. When the Echo plugin is inserted on a Mono Channel, the Left Channel and Right Channel sections are replaced with a single "Echo" section with only the Delay Time, Feedback Delay, and Feedback controls.

  Delay Time: Global Delay time for the Left Channel.

  Feedback Delay: Echo Delay time for the Left Channel.

  Feedback: Feedback percentage of the Left channel back to itself.

  L > R Feedback: Percentage of Left feedback signal which feeds back to Right Channel.

- **Right Channel:** Parameters that independently affect delay on the Right Channel.

  Delay Time: Global Delay time for the Right Channel.

  Feedback Delay: Echo Delay time for the Right Channel.

  Feedback: Feedback percentage of the Right channel back to itself.

  R > L Feedback: Percentage of Right feedback signal which feeds back to Left Channel.

- **Output:** Controls for adjusting the final output from this plugin.

  Dry/Wet (%): A percentage control of the output mix of “dry” or original signal to “wet” or processed signal. 0 is completely dry, 100% is completely wet.

  Output Level (dB): Adjusts the overall output level of the affected sound.

EQ - Fairlight EQ

The Fairlight EQ is a 6-band parametric equalizer with controls for boosting or attenuating different frequencies.

Each band has controls for filter type, including Bell, Lo-Shelf, Hi-Shelf, Notch, Frequency, Gain, and Q-factor (sharpness of the band).

Fairlight EQ plugin

Master Controls

The Equalizer window has the following controls located in the upper section of the interface

- **Preset Menu:** Your EQ settings can be saved as presets by clicking the Plus button to the left of the Preset menu and recalled by clicking the menu and making a selection. You can also scroll through the presets by clicking the left or right “arrow” (< or >) to the right of the drop-down menu.

TIP: You can save an EQ setting as the default by selecting the Set as Default option from the Options menu (three dots) located in the upper-right corner, to the right of the Reset button.

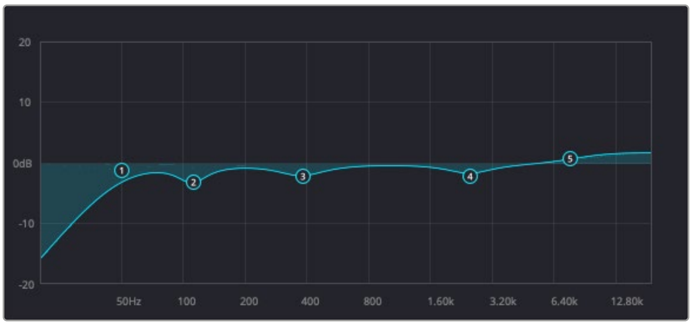
Fairlight EQ Presets can be copied and pasted to other instances of the plugin for a faster workflow.

- **Compare Buttons (A and B):** These buttons let you quickly compare two different EQ settings.

  If you create an EQ setting or select a preset when button A is active (red), then click button B and create or choose a different setting, clicking between the buttons quickly switches from one setting to the other.
- **Bypass Button:** Switched the Fairlight EQ on and off.
- **Reset Button:** Clicking the circular Reset icon, located in the upper-right corner of the EQ, undoes the most recent action you've taken in the plugin interface.

EQ Graph

This graph displays the current EQ curve with handles representing each EQ band. Although their position on the graph is based on the parameter settings on the plugin interface, you can drag any numbered handle to boost or attenuate the range of frequencies governed by an active band, based on the currently selected equalization filter.



Fairlight EQ graphical display

Bands 1 and 6

These parameters let you adjust the lowest and highest frequencies of the audio signal.

Band enable button: Switches each EQ band on and off.

Band filter type: Bands 1 and 6 can be switched between five filtering options as described below:

- **For Band 1, these include (from top to bottom):** Lo-Shelf, Notch, Hi-Shelf, and Hi-Pass.
- **For Band 6, the options are (from top to bottom):** Lo-Pass, Bell, Notch, and Hi-Shelf.

Frequency: Adjusts the center frequency of the EQ adjustment.

Q Factor: This parameter adjusts the width of affected frequencies and appears whenever the Bell option is selected. Lower values cover a wider range of frequencies; higher values narrow the frequency range.

This parameter does not affect the audio signal when either the Hi-Pass, Lo-Shelf, or Hi-Shelf filter is in use.

FAIRLIGHT

Bands 2 through 5

These bands included the following controls and parameters:

Band enable button: Turns each band of EQ on and off.

Band filter type: These Bands offer four filtering options (from top to bottom): Lo-Shelf, Bell, Notch, and Hi-Shelf.

Frequency: Adjusts the center frequency of the EQ adjustment.

Gain: Negative values attenuate the selected frequency, while positive values boost it.

This parameter does not affect the audio signal when the Notch filter is in use.

Q Factor: This parameter appears when the Bell setting is selected. It adjusts the width of affected frequencies. Lower values cover a wider range of frequencies; higher values narrow the frequency range.

This parameter does not affect the audio signal when either the Lo-Shelf, Notch, or Hi-Shelf filter is in use.

Flanger

An effect plugin, giving that unmistakable Flanger sound dating from the days of dual tape machines with a slight delay added to one in periodic intervals causing flanging as they got back in sync with one another. Typically used to add a sort of warbling harmonic interest to a signal, in a wide variety of ways.

An animated graph shows the results of adjusting the Modulation parameters of this plugin, giving you a visualization of the kind of warble that will be added to the signal as you make adjustments.

The Flanger Fairlight FX

FAIRLIGHT

The Flanger has the following controls:

- **Bypass:** Toggles this plugin on and off.
- **Input mode:** (Only visible when the Flanger is inserted on a multi-channel track.) Lets you choose how multiple channels are input to the Flanger. Stereo sets separate Left and Right channels. Mono sums Left and Right to both channels. Left inputs the Left channel only, and Right inputs the Right channel only.
- **Modulation:** A low frequency oscillator (LFO) used to drive the Flanger effect.

  Waveform (Hz): Specifies the shape of the LFO that modulates the rate of the Flanger. The three choices are Sine (a smooth change in rate), Triangle (a jerky change in rate), and Sawtooth (an abrupt change in rage). Affects the timing of the warble that is added to the sound.

  Rate: Speed of the LFO, affects the speed of the warble that is added to the sound. Low rate values create a slow warble, while high rate values create more of a buzz.

  Depth: Affects the "length" of the warble that is added to the sound. Low values add only the very beginning of a warble, high values add more fully developed warble.

- **Width:** Consists of a single parameter, Expansion, which sets Left/Right channel length differences, along with the phase offset of modulators.
- **Feedback:** These controls determine, in large part, how extreme the Flanging effect will be.

  Amount (%): The percentage of signal fed back to the Delay Line. Values can be positive or negative, the default is 0 (no effect). Increasing this parameter adds more of the Flange effect to the signal, lowering this parameter adds more of the inverted Flange effect to the signal. At values closer to 0, only a faint phase shift can be heard in the audio, but at values farther away from 0 (maxing at +/- 99), a gradually increasing warble becomes audible. The type of warble depends on the Modulation controls.

  High Ratio: Determines the attenuation of the echo over time.

- **Output:** Controls for adjusting the final output from this plugin.

  Dry/Wet (%): A percentage control of the output mix of "dry" or original signal to "wet" or processed signal. 0 is completely dry, 100% is completely wet.

  Output Level (dB): Adjusts the overall output level of the affected sound.

Foley Sampler

The Foley Sampler is a built-in sampler that makes it easy to add sound effects that you want to play using a keyboard, pad, or other MIDI performance device connected to your computer, to add timed sound effects to sync with onscreen visuals. This plugin has been designed to simplify the process of recording performed audio cues on the current track to which the sampler has been added.

Setting Up the Foley Sampler

Using the Foley Sampler to record samples played with a MIDI controller is easy.

1. Create an audio track for your sound effects or instrument recording.
2. Drag the Foley Sampler onto the track header to assign it to that track. The Foley Sampler window automatically appears. The Fairlight page knows this is an instrument with no inputs to the plugin, so this effect is automatically patched to that track's input, ready for recording.
3. If you have a MIDI controller of some kind connected to your computer and properly configured, it will appear in the MIDI dropdown menu at the upper-right corner of the Foley Sampler window (next to the Keyboard button). Choose your device from this menu, and the Keyboard button will highlight to show it's enabled.

At this point, the Foley Sampler is ready to be used, but by default it has no samples loaded to play. The next step is to add sound effects.

Adding Sound Effects

You can add prerecorded sound effects or instruments to the Foley Sampler in the following ways:

- Drag and drop an audio file in a supported format from the file system onto the Foley Sampler window
- Drag and drop a sound effect from the Sound Library onto the Foley Sampler window
- Click the Foley Sampler window's Option menu and input sounds from your attached drives.

Once you've loaded a sound effect of some kind, it's automatically mapped to the top four keys of your instrument, starting at C2. Pressing a key or pad of your controller will play that sound.

To see the sound's waveform, click the Sample button at the top of the Sound panel. In this example, there are a series of continuous footsteps recorded in a row that we can use.

A footsteps recording loaded into the Foley Sampler

Splitting Sound Effects

It's common to use pre-recorded library sound effects that consist of a series of recorded footsteps, cloth rustles, punches, or other "foleyed" sound recordings, in order to play variations of a specific kind of repetitive sound effect in sync to action that's happening on screen. The Foley Sampler lets you do this easily.

1. After you've loaded a sound effect, click the Foley Sampler window's Option menu and choose Split Sample to automatically split the current sample into slices based on an analysis of its noise floor, and assign each slice to a set of keys or pads on your selected MIDI device. Continuing with the previous example, each footstep has been split and assigned to a different key.

2 To adjust the timing of each slice of the sound effect that's been split apart, you can click on the assignment text above the keyboard at the bottom of the window to see that slice in the Sample view.

3 You can adjust the Range Start and Range End parameters to encompass as much or as little of that slice as you want to play back.

4 If you want a sample to loop if a note is held down, you can enable the Loop button, and then adjust the Loop Start and Loop End parameters to choose how much of each slice will loop.

5 To delete slices that aren't useful, you can select a slice you don't like and press Shift-Delete to clear that slice from the virtual keyboard.

A single slice of the footsteps recording after being split in the Foley Sampler, with the range of the effect adjusted

Assigning Sound Effects Manually

When you first load sound effects into the Foley Sampler, they're automatically assigned to a series of notes. Each additional sound effect that you load is automatically assigned to the next series of notes to the right. Once all notes are occupied, additional sound effects will shuffle all previous assignments to the left.

You also have the option of manually assigning sound effects that you load. This is useful when you want to manually load a variety of different sound effects all at once (such as a combination of punch sounds, human grunts, and cloth rustles to use in a fight scene), and assign them to particular notes of your choosing.

1 Load a sound effect you want to map.

2 Click the Mapping button to remap the range of notes it corresponds to.

3 Use the Low and High parameters to select a range of notes for the selected sound.

4 You also have the option of tuning the pitch of specific sound effects, if necessary.

The Mapping controls used to assign a sound to a particular set of notes

Adjusting Sound Effects

If you want to customize a sample or slice further, you can select it above the virtual keyboard and use the controls on the Level panel to control the dynamics of audio playback, or you can use the controls on the Filter panel to EQ the sound.

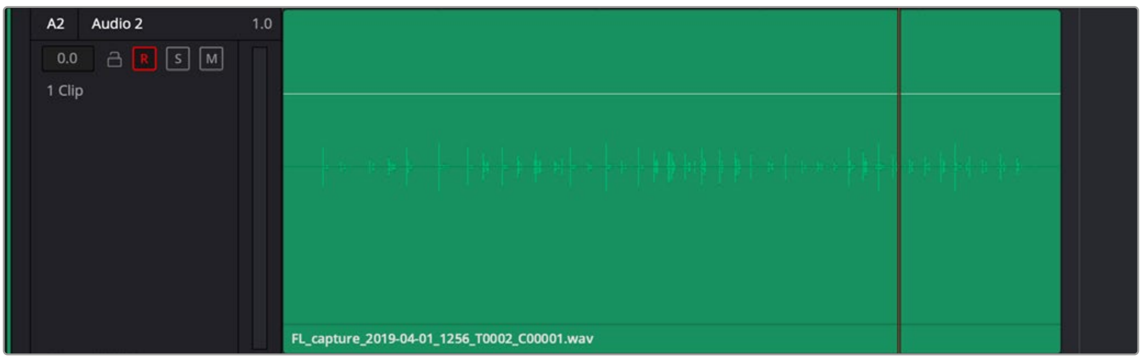
TIP: If a sample or slice is set to loop, you can press the Control key and click a key on the virtual keyboard to initiate looping, so you can hear these adjustments as you're making them.

Playing and Recording Sound Effects

Once you've set up the Foley Sampler with sound effects you can play from a MIDI controller, recording those sound effects is simple.

1 Make sure that the "Save clips to" field in the Capture and Playback panel of the Project Settings is correctly set up to record to the desired storage volume, using the Browse button, if necessary.
2 Click the R button in the track header of the audio track to which you applied the Foley Sampler, to put that track into Record Enable mode.
3 Click the Record button in the Fairlight toolbar.
4 While Fairlight records, use the keys or buttons of your MIDI controller to play sounds in sync to the picture on your display. When you're done, click the Stop button.

You should now have a recorded clip containing the sound effects you played, in sync to the picture. If they're a bit out of sync, you can always use the Elastic Wave audio retiming controls to tighten the sync without re-recording everything.



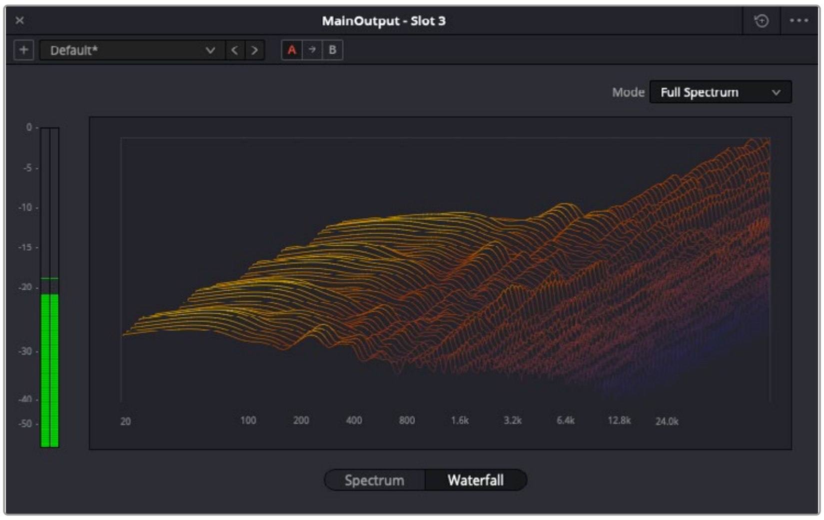
A new clip of audio created by recording sounds played via the Foley Sampler

Frequency Analyzer

The Frequency Analyzer provides visual information about your clips or tracks as either a Spectrum or Waterfall graph to identify possible sonic issues so you can further sculpt your audio.

Options in the Mode menu in the upper right let you examine either the Full frequency range of the signal (default) or close-up views of the Low, Midrange, or High frequencies.

Frequency Analyzer Spectrum view



Frequency Analyzer plugin Waterfall view

Gain

This plugin is used to accurately increase or decrease the volume level of an audio signal by a specific number of decibels. It is helpful in numerous contexts, such as making a quick adjustment to the overall volume of a track that has extensive mixer or plugin automation written to it.

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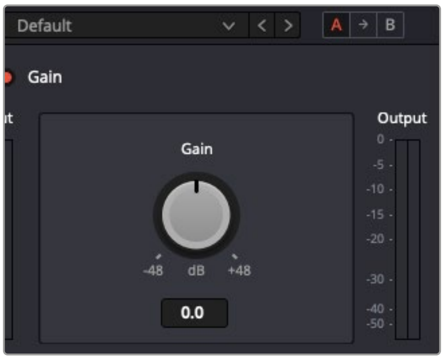
FAIRLIGHT



The Fairlight Gain plugin

LFE Filter

A low-pass filter that you can apply to a FlexBus or Main that's in a surround sound format, to feed low-frequency sound to an LFE channel as part of a surround sound mix. The filter will exclude any sounds above your chosen frequency setting to the audio that is sent to a sub-woofer in the LFE channel. It helps keep unwanted and unnecessary audio from being sent to the sub-woofer and increases playback clarity.

A Frequency control lets you choose which low frequency range you want to include, and a Trim control lets you set the level of the resulting LFE channel. If it is multi-channel but has no LFE channel available, such as a 5.0 format, then this plugin does nothing.

The LFE plugin

Limiter

The Limiter detects and reduces peaks in an audio signal, so the overall level can be boosted without clipping while reducing the dynamic range in a similar manner to a compressor.

This is a true peak limiter that can look 64 samples ahead of the input signal, allowing an extremely smooth limiting response.

The Input control lets you adjust the incoming signal level, while the Threshold and Release control when limiting occurs and for how long. The graph indicates which parts of the signal are being affected.

The Limiter plugin

Limiter controls

Bypass: Toggles this plugin on and off.

Input Meter: Shows the level of the incoming signal.

Input Level: This control lets you adjust the incoming signal level within a range of -18 dB to +18 dB. Activating the Soft button causes the limiter to use a gentler attack.

Ceiling: This determines the threshold (level) at which the Limiter starts reducing the input signal. When set to -24 dB, any signal above that level is limited. When set to 0 dB, no limiting occurs.

Release: This determines how long the limiting (signal reduction) takes place. Setting this control to 0.01 milliseconds results in a very fast release time, while the longer release time of 1000 milliseconds is the slowest.

Reduction Meter: This shows the amount of level (gain) reduction applied to the input signal.

Output Meter: Shows the output level after the limit.

Limiter Sidechaining

The Fairlight FX Limiter supports sidechaining, which lets you temporarily lower the level of a target track or bus using the audible signal from a source track or bus.

This is very helpful when multiple tracks are “fighting” for space in a mix because they share some or all the same frequency range. For example, a track or bus with dialogue and a bus or track with traffic and crowd noise.

Essentially, this is another form of “ducking” where, in the example above, the dialogue attenuates the traffic and crowd noise until the dialogue stops. This allows the voices to be heard more prominently in the mix.

Sidechain controls

FAIRLIGHT

Using the Limiter Sidechain

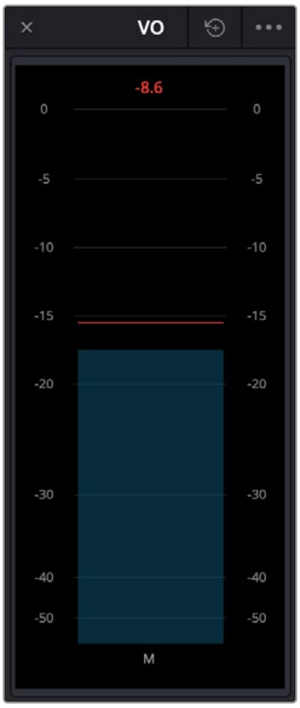
Carrying on with the example mentioned above:

1 Open the Limiter on the destination track or bus with the traffic and crowd noise.
2 Select the Dialogue track or bus in the source dropdown to the right of the Listen button.
  - This is the audio you're using to attenuate the level of the target track.
  - Using a bus as the source lets you use multiple sources to trigger changes in your target track.

Clicking the Listen button lets you hear the incoming signal if you need to confirm that the correct input signal is coming into the Limiter. This button turns yellow when active.

Meter

A sample peak processing meter that's useful for temporarily adding a meter to a specific track or FlexBus. These meters are useful for instances where you want a large meter that focuses on a specific bus or track while you're working.



The Meter Fairlight FX

These meters are presented very simply, with a gray bar indicating level and a red peak line that holds for two seconds, which indicates the highest peak. A numeric reading at the top of the meter gives the exact level, in dB. This number continues to hold, indicating the loudest level measured for any given stretch of playback.

Meter can be resized by pulling from the bottom right and has the following controls located in the option menu:

Reset Peak on Play: When enabled, the numeric peak level is reset every time playback stops and starts again. When disabled, the numeric peak level persists until changed by a higher peak.

Reset: Resets the numeric peak level.

FAIRLIGHT

Modulation

An effect plugin. General purpose modulation plugin for sound effects or sound design. Four effects combine an LFO, FM adjustment, AM adjustment, Sweep and Gain filters to allow simultaneous frequency, amplitude and space modulation. In conjunction with the Rotation controls, simple Tremelo and Vibrato effects can be combined with auto-filter and auto-Pan tools in order to provide texture and movement to a sound.

An animated graph shows the results of adjusting the Modulator, Frequency, and Amplitude parameters of this plugin, giving you a visualization of the kind of modulations that will be applied to the signal as you make adjustments. Output meters let you see what level is being output.

The Modulation Fairlight FX

Modulation has the following controls:

- **Bypass:** Toggles this plugin on and off.
- **Modulator:** A low frequency oscillator (LFO), shown in blue in the animated graph.

  Shape: Specifies the shape of the LFO waveform that modulates the audio. Six options include Sine, Triangle, Saw1, Saw2, Square, Random.

  Rate (Hz): Adjust the speed of the modulating LFO. Lower settings result in warbling audio, while extremely high settings result in buzzing audio the timbre of which is dictated by the shape you've selected.

- **Frequency:** Frequency modulation (FM) of a secondary oscillator, shown as green in the animated graph.

  Level (%): Acts as a Dry/Wet knob controlling the amount of Frequency Modulation that's applied, intensifying or easing off the effect.

  Phase: Since each of the four primary effects within this plugin can be applied together, along with the fact that modulation with level components (Tremelo/Rotation/Filter) have the ability to combine or cancel out one another, phase controls are available. Altering the phase of an individual effect allows control of such interaction (e.g., cancel out a high level change, or offset a cancellation).

- **Filter:** Sweep and gain filters.

  Filter (%): Acts as a Dry/Wet knob controlling the amount of filter sweep and gain to additionally use to modify the signal. The amount you've selected is previewed in a 1D graph to the side.

  Tone: Adjusts the center frequency of sweep.

  Phase: Since each of the four primary effects within this plugin can be applied together, along with the fact that modulation with level components (Tremelo/Rotation/Filter) have the ability to combine or cancel out one another, phase controls are available. Altering the phase of an individual effect allows control of such interaction (e.g., cancel out a high level change, or offset a cancellation).

- **Amplitude:** Amplitude modulation (AM) of a secondary oscillator, shown as green in the animated graph.

  Level (%): Acts as a Dry/Wet knob controlling the amount of Amplitude modulation applied. (Disabled in Ring Modulation Mode.)

  Phase: Since each of the four primary effects within this plugin can be applied together, along with the fact that modulation with level components (Tremelo/Rotation/Filter) have the ability to combine or cancel out one another, phase controls are available. Altering the phase of an individual effect allows control of such interaction (e.g., cancel out a high level change, or offset a cancellation).

  Ring Modulation Mode: Enables a Ring Modulation effect (where the signal is multiplied by the modulator, rather than modulated by it).

- **Rotation:** These controls are only available when applied to a multi-channel track.

  Rotate: Amount of Rotation applied.

  Offset: Start offset of rotation in order to further place the signal in space.

  Phase: Since each of the four primary effects within this plugin can be applied together, along with the fact that modulation with level components (Tremelo/Rotation/Filter) have the ability to combine or cancel out one another, phase controls are available. Altering the phase of an individual effect allows control of such interaction (e.g., cancel out a high level change, or offset a cancellation).

- **Output:** Controls for adjusting the final output from this plugin.

  Dry/Wet (%): A percentage control of the output mix of "dry" or original signal to "wet" or processed signal. 0 is completely dry, 100% is completely wet.

  Output Level (dB): Adjusts the overall output level of the affected sound.

Multiband Compressor

A dynamics processor that compresses in highly definable frequency bands. The graph displays frequencies in Hertz horizontally and gain in decibels vertically. This allows for precise compression specific to defined frequency bands and is useful for taming only one or several parts of a signal.

The default setting of the Multiband Compressor

Each band has dynamics control in the determined frequency ranges of Low, Med, and High and are adjustable by pushing the intersection mark on the graph per band or by typing in a frequency.

Bands 1 – 4: Controls for each band.

- **Threshold:** Sets the maximum allowable output level. The default is –25dB. The range is from –50 to 0dB.
- **Gain:** Can add or attenuate up to 12 dB of gain.
- **Ratio:** Adjusts the compression ratio. This sets the gain reduction ratio (input to output) applied to signals that rise above the threshold level. The default is 1.5.0:1. The range is 1.0:1 to 7:1.
- **Limit:** Limits the output amount by up to 15 dB. The default is 4.5 dB.
- **Attack:** Adjusts the attack rate time constant of the sidechain detector. The default is 1.4 mS. The range is 0.1 to 100 mS.
- **Hold:** Keeps dynamics from being triggered again until a certain amount of time has passed, in mS (milliseconds). Defaults to 0mS. The range is from 0 to 4000 mS.
- **Release:** Adjusts how quickly the sidechain detector stops applying dynamics when a signal goes back below the threshold. The default is 150 mS. The range is 50 mS to 4.0 S.

Master: Controls for adjusting the final output from this plugin.

- **Gain (dB):** Adjusts the overall output level of the affected sound by adding or reducing 18 dB of gain.
- **Q:** Adjusts the width of affected frequencies. Lower values include a wider range of frequencies, higher values include a narrower range of frequencies.

Noise Reduction

A repair plugin designed to reduce a wide variety of noise in all kinds of recordings. A graph shows a spectral analysis of the audio being targeted, along with a purple overlay that shows what noise is being targeted. Two audio meters let you evaluate the input level (to the left) versus the output level (to the right), to compare how much signal is being lost to noise reduction. There are three default presets: De-Hiss, De-Rumble, and De-Rumble and Hiss.

The Noise Reduction Fairlight FX in action

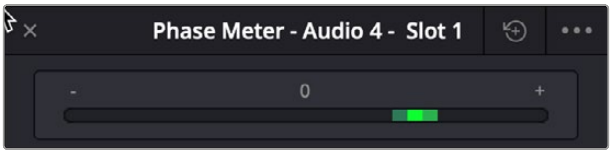
Noise Reduction has the following controls:

- **Bypass:** Toggles this plugin on and off.
- **Listen to Noise Only:** This checkbox at the top right lets you listen only to the noise that is being removed. This is very useful to determine if too much signal is being removed or if more noise attenuation can be applied.
- **Threshold (in dB):** Relates to the signal-to-noise ratio (SNR) in the source recording. Recordings with a poor signal-to-noise ratio will require a higher threshold value, resulting in more noise reduction being applied.
- **Attack (in ms):** Primarily useful in Auto Speech mode, this controls the duration over which the noise profile is detected. Ideally, the attack time should match the variability of the unwanted noise. A low value corresponds to a faster update rate of the noise profile and is useful for quickly varying noise; a high value corresponds to a slower update rate and can be used for noise that's more consistent.
- **Sensitivity:** Higher sensitivity values exaggerate the detected noise profile; the result is that more noise will be removed, but more of the dialogue you want to keep may be affected.
- **Ratio:** Controls the attack time of the signal profile relative to the attack time of the noise profile. A faster ratio detects and preserves transients in speech more easily, but the resulting speech profile is less accurate.
- **Frequency Smoothing:** Smooths the resulting signal in the frequency domain to compensate for harmonic ringing in the signal after the noise has been extracted.
- **Time Smoothing:** A toggle button enables smoothing of the resulting signal in the time domain as well.
- **Dry/Wet:** A percentage control of the output mix of "dry" or original signal to "wet" or processed signal. 0 is completely dry, 100% is completely wet.
- **Level:** To let you compensate for level that may be lost due to the noise reduction operation you're applying, this applies a pre-gain in, from -6dB to +18dB, just before the dry/processed mix.

Phase Meter

Phase cancellation is a phenomena where the waveforms of a stereo recording (for example a stereo recording of a music performance) go slightly out of sync with one another for whatever reason, and begin to cancel one another out in unpredictable ways, resulting in the audio sounding strange. This results in poor quality audio and can cause problems when you're trying to compress a mix to a distribution format such as AAF or MP3.

The Phase Meter plugin is a visual meter that lets you evaluate whether or not a signal is in phase and is meant to be applied to a bus so you may evaluate the phase of a mix and correct whatever problems may be occurring. The position of a green dot within a horizontal meter indicates the phase of the signal. When there's no signal or a signal on only one half of a stereo bus, the dot appears in the center (0). When the signal is out of phase, the dot appears all the way to the left (–). When the signal is in phase, the dot appears all the way to the right (+).



The Phase Meter plugin

Pitch

An effects plugin that shifts audio pitch without altering clip speed.

The Pitch Fairlight FX

Pitch has the following controls:

- **Bypass:** Toggles this plugin on and off.
- **Semitones:** A "coarse" adjustment that can shift audio pitch up to +/- 12 semitones.
- **Cents:** A "fine" adjustment that can tune audio pitch in +/-100th of a semitone.
- **Dry/Wet:** A percentage control of the output mix of "dry" or original signal to "wet" or processed signal. 0 is completely dry, 100% is completely wet.

Reverb

A spatial simulation plugin, capable of recreating multichannel reverberation corresponding to rooms of different sizes, adjustable via a graphical 3D cube control. This plugin lets you take a "dry" recording and make it sound as if it's within a grand cathedral, an empty room, a tiled bathroom, or other spaces.

To understand this plugin's controls, it helps to know that the signal follows three paths which are combined to create the final effect:

- A direct path.
- An early reflection path (ER) simulating early reflection rays obtained from the first multiple reflections on the walls, traveling from the virtual source to the virtual listener.
- A late reverberation path (Reverb) simulating the behavior of an acoustic model of the room.

A graph shows an approximate visualization of the reverb's effect on the frequencies of the audio signal.

The Reverb Fairlight FX

Reverb has the following controls:

- **Bypass:** Toggles this plugin on and off.
- **Room Dimensions:** By controlling the size of the virtual room a sound is to inhabit, these parameters simultaneously control the configuration of Early Reflection and Late Reverberation processing. The acoustic modes from this simulated room are computed and fed to Late Reverberation processing. The shape, gain, and delay of the first reflections are computed and then fed to Early Reflection processing.

  Height, Length, Width: Defines the dimensions of the reverberant space, in meters.

  Room Size: The calculated Room Width x Length, in meters.

- **Reverb:** Additional controls that further customize the configuration of Early Reflection and Late Reverberation processing.

  Pre Delay: Increase or negate the propagation time from the virtual source to the virtual listener. As a result, it modifies the initial delay time between the source signal and the first reflection.

  Reverb Time: Decay time of the Reverb tail. It controls the overall decay time of the acoustic modes from late reverberation processing.

  Distance: Modifies the distance between the virtual source and the virtual listener. It modifies only the configuration of early reflections processing.

  Brightness: Modulate the shape of the decay time over frequency. At maximum brightness, decay time is identical at any frequency. At minimum brightness, higher frequencies result in shorter decay time and therefore duller sound.

  Modulation: Adds random low-frequency phase modulation from the tapping point of ER processing. At 0%, modulation is not used.

- **Early Reflection Tone:** Four post equalization controls modify the tone of early reflections to suit a particular room's characteristics.

  Low Gain: Amount of gain added to the low frequency.

  Low Frequency: Frequency range of 150 Hz to 500 Hz.

  High Gain: Amount of gain added to the high frequency.

  High Frequency: Frequency range of 1k Hz to 16k Hz.

- **Reverb Tone:** Four post equalization controls modify the tone of the reverb tail to suit a particular room's characteristics.

  Reverb Tail Low Gain: Amount of gain added to the low frequency.

  Reverb Tail Low Frequency: Frequency range of 150 Hz to 500 Hz.

  Reverb Tail High Gain: Amount of gain added to the high frequency.

  Reverb Tail High Frequency: Frequency range of 1k Hz to 16k Hz.

- **Output:** These controls recombine the three audio processing paths into a single output signal.

  Dry/Wet: A percentage control of the output mix of "dry" or original signal to "wet" or processed signal. 0 is completely dry, 100% is completely wet.

  Direct Level: The amount of the direct level to mix into the final signal.

  Early Reflection Level: The amount of early reflection to mix into the final signal.

  Reverb Level: The amount of reverb to mix into the final signal.

Soft Clipper

The Soft Clipper is a limiting processor that reduces the output level above a defined threshold in a rounded manner so that peaks are more cleanly attenuated. The Soft Clipper plugin will impart saturation effects when pushed hard above the threshold, allowing for the introduction of warmth and subtle distortion to the sound. A graph shows the shape of the curve adjustment this plugin makes to the audio.

A soft clipper is often combined with a standard limiter in order to increase perceptual loudness of material without imparting harshness.

The Soft Clipper Fairlight FX

Threshold: Introduces input gain to the signal prior to hitting the clipper, forcing audio peaks over the threshold by that amount. As such, it will drive the saturation and distortion.

Shape: The shape of the clipper can be varied to change the character of the soft clipper from full soft-clipping (all the way right, where the peaks are rounded) to full hard-clipping (all the way left, where the peaks are squared off).

Output Level: Lets you adjust the output gain to compensate for signal loss during soft clipping, if necessary.

Stereo Fixer

A simple plugin designed to fix stereo source material in cases where only one side of a stereo signal was recorded, where one side of a stereo recording is a different level to the other, or where the stereo channels have been incorrectly Left/Right swapped.

This plugin can also be used as a "Mid/Side" decoder, for recordings that were made using this microphone technique.

This plugin is for stereo clips only.

The Stereo Fixer Fairlight FX

- **Format:** The input processing mode you want to use to fix the stereo output.

  Stereo: (Default) No format conversion is performed.

  Reverse Stereo: Swaps the Left and Right side.

  Mono: The output from the plugin is a mono mix of the two inputs.

  Left Only: The left input is sent to both left and right outputs.

  Right Only: The right input is sent to both left and right outputs.

  M/S: The left output is the left (Mid) input minus the right input (Side). The right output is the left (Mid) input plus right input (Side).

- **Left/Right Gain:** Lets you apply independent gain on the left or right outputs. This gain is applied after (post) the input processing mode.

FAIRLIGHT

> **TIP:** For a comprehensive M/S decoder solution, simply chain two Stereo Fixer plugins together. Use the first unit to control the Side signal level, thus controlling the width of the second unit (set to M/S).

Stereo Width

An enhancement plugin that increases or reduces the spread of a stereo signal in order to widen or reduce the separation between channels. If this plugin is added to a Mono channel, it will be disabled, as there is no stereo width to either distribute or control.

A graph shows the currently selected width of stereo distribution as a purple arc, while inside of that graph a stereo meter shows the Left and Right distribution of the audio signal. Two audio meters measure levels, an Input meter to the left, and an Output meter to the right.

The Stereo Width Fairlight FX in action

Stereo Width has the following controls:

Width: Lets you control the spread of the stereo output. Settings range from 0 (Mono) to 1 (Stereo) to 2 (extra wide stereo).

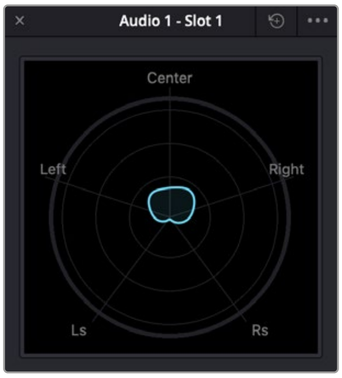
Diffusion: Adds more complexity to the output.

Sparkle: Adds more high frequencies to the spread.

Surround Analyzer

The Surround Analyzer is a graphical meter that shows a spatial image of the audio being measured, rather than a typical bar graph meter. Due to its changing shape because of the signal being played, sometimes it is referred to as the "jellyfish meter."

This type of metering is very useful; rather than relying on bars to indicate the directions in which audio is radiating, you can clearly see the relationship of all of the channels to one another.



The signal here is radiating more to the right, indicating the panning of the audio.

Vocal Channel

An enhancement plugin for general purpose vocal processing consisting of High Pass filtering, EQ, and Compressor controls.

Side by side EQ and Dynamics graphs are presented above the controls. An output audio meter lets you monitor the final signal being produced by this plugin.

The Vocal Channel Fairlight FX

Vocal Channel has the following controls:

- **High Pass:** Enabled by a toggle, off by default. Has a single frequency knob that sets the threshold below which frequencies are attenuated to reduce boominess or rumble.
- **EQ:** A three-band EQ for fine tuning the various frequencies of speech, enabled by a toggle, including Low, Mid, and High Mode, Frequency, and Gain controls

  Low/Mid/Hi Mode: Lets you choose from different filtering options to use for isolating a range of frequencies to adjust. Different bands present different options.

FAIRLIGHT

Low/Mid/Hi Freq (Hz): Lets you choose the center frequency to adjust.

Low/Mid/Hi Gain (dB): Lets you boost or attenuate the selected frequencies.

- **Compressor:**

  Threshold (dB): Sets the signal level below which compression occurs. Defaults to -25dB. The range is from -40 to 0dB.

  Reaction: Adjusts how quickly compression is applied when a signal exceeds the threshold. The default is 0.10.

  Ratio: Adjusts the compression ratio. This sets the gain reduction ratio (input to output) applied to signals which rise above the threshold level. The default is 1.5:1. The range is 1.1 to 7.0.

  Gain (dB): Lets you adjust the output gain to compensate for signal loss during compression, if necessary.